Master of Engineering - M.E in Computer Science and Engineering (Artificial Intelligence and Machine Learning)

Syllabus

July 2026 Onwards

MANIPAL SCHOOL OF INFORMATION SCIENCES,

MANIPAL ACADEMY OF HIGHER EDUCATION,

MANIPAL – 576104, KARNATAKA, INDIA.

Program Educational Objectives / Outcomes (PEOs)

PEO1: Acquire solid mathematical and computational skills essential for understanding, applying, developing, and deploying modern artificial intelligence and machine learning algorithms.

PEO2: Produce industry-ready graduates with practical experience in structuring artificial intelligence and machine learning projects using state-of-the-art software.

PEO3: Address multi-disciplinary challenges through coursework and projects by adapting to the rapidly advancing developments in artificial intelligence and machine learning approaches.

Program Objectives / Outcomes (POs)

PO1: An ability to independently carry out research/investigation and development work to solve practical problems.

PO2: An ability to write and present a substantial technical report/document.

PO3: Demonstrate a degree of mastery over the area as per the specialization of the program which should be at a level higher than the requirements in an appropriate bachelor program.

PO4: Identify appropriate and efficient algorithmic approaches for solving real-life challenges using artificial intelligence and machine learning principles and state-of-the art software prevalent in industry and academia.

PO5: Develop teamwork and leadership skills for addressing challenges of social importance for sustainable societal development using ethical artificial intelligence and machine learning approaches.

Program Structure

Semester I										
Course Code	Course Name	No. of Hrs./week				Duration of Exam in Hrs	Maximum Marks		
		Lecture	Tutorial	Practical	Credit		Internal 50	External 50	Total 100
AME 5002	Applied Machine Learning	3	-	-	3	3	50	50	100
AME 5003	Principles of Natural Language Processing	3	-	-	3	3	50	50	100
AME 5101	Applied Linear Algebra	3	-	-	3	3	50	50	100
AME 5102	Applied Probability and Statistics	3	-	-	3	3	50	50	100
BDE 5101	Algorithms and Data Structures for Big Data	3	-	-	3	3	50	50	100
AME 5052	Applied Machine Learning Lab	-	-	3	1	3	50	50	100
AME 5053	Principles of Natural Language Processing Lab	-	-	3	1	3	50	50	100
AME 5151	Applied Linear Algebra Lab	-	-	3	1	3	50	50	100
AME 5152	Applied Probability and Statistics Lab	-	-	3	1	3	50	50	100
BDE 5151	Algorithms and Data Structures for Big Data Lab	-	-	3	1	3	50	50	100
MPT 5100	Mini Project - I	-	-	-	4	-	100	-	100
PSD 5100	Professional Skill Development - I	-	-	-	1	-	100	-	100
Total		**15**	**-**	**15**	**25**				

Semester II									
Course Code	Course Name	No. of Hrs. / week				Duration of Exam in Hrs	Maximum Marks		
		Lecture	Tutorial	Practical	Credit		Internal 50	External 50	Total 100
AME 5202	Deep Learning	3	-	-	3	3	50	50	100
AME 5204	Applied Computer Vision	3	-	-	3	3	50	50	100
AME 5205	Advanced Generative AI and Agentic Systems	3	-	-	3	3	50	50	100
	Elective - I	3	-	-	3	3	50	50	100
	Elective - II	3	-	-	3	3	50	50	100
AME 5252	Deep Learning Lab	-	-	3	1	3	50	50	100
AME 5254	Applied Computer Vision Lab	-	-	3	1	3	50	50	100
AME 5255	Advanced Generative AI and Agentic Systems Lab	-	-	3	1	3	50	50	100
	Elective - I Lab	-	-	3	1	3	50	50	100
	Elective - II Lab	-	-	3	1	3	50	50	100
MPT 5200	Mini Project – II	-	-	-	4	-	100	-	100
PSD 5200	Professional Skill Development – II	-	-	-	1	-	100	-	100
TOTAL		**15**	**-**	**15**	**25**				

Semesters III & IV									
AME 6098	Project Work	-	-	-	30				
Total Number of Credits to Award Degree							80		

List of Electives (Theory)

Elective - I		Elective - II	
Course Code	Course Name	Course Code	Course Name
AME 5232	Reinforcement Learning	AME 5241	Applied Mathematics for Machine Learning
BDE 5001	Architecture of Big Data Systems	AME 5243	Advanced Applications of Probability and Statistics
CCE 5001	DevOps for Cloud	AME 5244	AI/ML on Cloud
ESE 5234	Entrepreneurship	ESE 5244	IT Project Management

List of Electives (Lab)

Elective - I		Elective - II	
Course Code	Course Name	Course Code	Course Name
AME 5282	Reinforcement Learning Lab	AME 5291	Applied Mathematics for Machine Learning Lab
BDE 5051	Architecture of Big Data Systems Lab	AME 5293	Advanced Applications of Probability and Statistics Lab
CCE 5051	DevOps for Cloud Lab	AME 5294	AI/ML on Cloud Lab
ESE 5284	Entrepreneurship Lab	ESE 5294	IT Project Management Lab

Semester I

AME 5002: Applied Machine Learning

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Apply different types of supervised machine learning algorithms to practical problems.

CO2: Apply different types of unsupervised machine learning algorithms to practical problems.

CO3: Analyse different types of machine learning paradigms.

Unit I **12 hrs**

Introduction to Machine Learning: Decision Trees, Linear Models: Overview of Supervised (regression and classification), unsupervised (clustering and dimensionality reduction), semi-supervised, and reinforcement learning with practical examples - Machine learning nomenclature: raw data, types of features and outputs, feature vector, Handling missing data, encoding categorical variables, feature scaling and normalization

Decision tree model of learning: Classification and regression using decision trees, splitting criteria, entropy, information gain, Gini impurity – Overfitting & Pruning in decision trees.

Unit II **12 hrs**

Linear regression: Model, metrics (MSE, RMSE, R-sqaure), estimation, and interpretation of coefficients, Introduction to bias/variance trade-off, regularization methods (Ridge and Lasso).

Logistic regression for binary classification: Sigmoid function, decision boundary, model interpretation.

Model evaluation and validation: Performance metrics such as accuracy, precision, recall, F1-score, ROC curve and AUC. Cross-validation techniques.

Nearest neighbours' algorithms - Probabilistic modelling: Nearest neighbours' algorithms - Probabilistic modelling of data using parameters, Introduction to maximum likelihood estimation (MLE) of parameters, Naive Bayes model for classification.

Unit III **12 hrs**

Ensemble learning methods: Bagging and Random Forest, Boosting techniques. Introduction to hyperparameter tuning and model improvement.

Clustering: K-means clustering - computing distances and similarities, limitations, Hierarchical clustering- dendrogram construction, types of linkage.

Dimension reduction: Principal component analysis (PCA)-concept, mathematics and applications.

Gaussian Mixture Models (GMM): probabilistic clustering, mixture of Gaussian distributions, expectation-maximization (EM) algorithm for parameter estimation.

References

1. G. James, D. Witten, T. Hastie, R. Tibshirani, and J. Taylor, An Introduction to Statistical Learning: with Applications in Python, Springer Texts in Statistics. Cham, Switzerland: Springer, 2023. Online resource available at https://www.statlearning.com/
2. L. G. Serrano, Grokking Machine Learning, S. Thrun, Foreword, Manning Publications, 2021. Online resource from Manning Publications available at https://www.manning.com/books/grokking-machine-learning
3. J. VanderPlas, Python Data Science Handbook, Sebastopol, CA: O'Reilly Media, 2016.
4. A Course in Machine Learning, Hal Daumé III – Online resource available at http://ciml.info/
5. Module: Introduction to Machine Learning (https://www.intel.com/content/www/us/en/developer/tools/oneapi/training/academic-program/educators/intro-machine-learning-training-kit.html).
6. Learning path: Understand data science for machine learning (https://learn.microsoft.com/en-us/training/paths/understand-machine-learning/).

AME 5003: Principles of Natural Language Processing

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Understand text processing for information extraction and retrieval.

CO2: Compare and contrast state-of-the-art word vectorization models for real-world problems.

CO3: Apply deep learning-based approaches for natural language processing tasks.

Unit I **10 hrs**

Introduction to Text Processing: Information extraction using regular expressions – Stages of text processing: tokenization, normalization, stemming, lemmetization and stop words-removal.

Basic linguistic processing: Part-of-Speech (POS) tagging and Named Entity Recognition (NER).

Information Retrieval: term document incidence matrices, inverted index construction and query processing, basic ranking using TF-IDF.

Unit II **12 hrs**

Language Modelling, Sentiment Analysis, Word Embeddings: Language modelling: N-grams, N-gram probabilities, evaluation and perplexity. Sentiment Analysis using Naïve Bayes.

Text representation models: Bag-of-Words, TF-IDF vector model, cosine similarity. Word embeddings: Neural word embeddings – Word2Vec (CBOW and Skip-gram) and GloVe models

Unit III **14 hrs**

NLP with Deep Learning, Applications of Natural Language Processing: Neural approaches for NLP: Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks for sequence modelling, GRU (Gated Recurrent Unit) Attention mechanism and sequence-to-sequence models.

Transformer architecture: Encoder and decoder transformers, self-attention mechanism.

Pre-trained language models: BERT and DitilBERT fine-tuning for downstream NLP tasks.

Applications of NLP: Machine translation, question answering, chatbots, and text summarization.

References

1. IBM AI Engineering Professional Certificate (https://www.coursera.org/professional-certificates/ai-engineer).
2. S. Vajjala, B. Majumder, A. Gupta, and H. Surana, *Practical Natural Language Processing*. Sebastopol, CA, USA: O'Reilly Media, Jun. 2020.
3. D. Jurafsky and J. H. Martin, *Speech and Language Processing*, 3rd ed. (draft). [Online]. Available: https://web.stanford.edu/~jurafsky/slp3/
4. S. Bird, E. Klein, and E. Loper, *Natural Language Processing with Python: Analyzing Text with the Natural Language Toolkit*, 1st ed. London, U.K.: ISTE Ltd., 2017. [Online]. Available: https://www.nltk.org/book/
5. Y. Goldberg, *A Primer on Neural Network Models for Natural Language Processing*. [Online]. Available: http://faculty.cse.tamu.edu/huangrh/Spring18/nnlp.pdf
6. D. Rao and B. McMahan, *Natural Language Processing with PyTorch*, 1st ed. Sebastopol, CA, USA: O'Reilly Media, 2019.

AME 5101: Applied Linear Algebra

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Apply vector operations for similarity learning.

CO2: Analyse matrix representations in AI & ML models.

CO3: Apply matrix calculus for optimization in learning systems.

Unit I **10 hrs**

Vector Operations & Geometry in AI Systems

Algebraic & geometric view of vectors; Dot product & similarity; Norms & regularization; Cosine similarity; Distance metrics; Standardization vs normalization; High-dimensional embeddings; Vector similarity in NLP; Feature representations; Vectorized computation.

Unit II **14 hrs**

Matrix Representations & Transformations: Matrix operations; Matrix-vector product; Matrix-matrix product; Forward propagation in deep learning; Batch computation; Elementary row operations and RREF; Solving systems of linear equations; Linear dependence and independence of vectors; Rank & subspaces; Matrix factorization: attention mechanism and low-rank representations; projections and dimensionality reduction using PCA.

Unit III **12 hrs**

Matrix Calculus for Learning Systems and visual: introduction to sensitivity & gradient in one dimension; Chain rule; Gradient in multiple dimensions; Matrix derivatives; Optimization using gradient descent; Batch vs stochastic learning; Backpropagation for binary and multiclass classification and regression.

References

1. Introduction to Applied Linear Algebra, Vectors, Matrices, and Least Squares, Stephen Boyd & Lieven Vandenberghe, Cambridge University Press, 1st Edition, 2018. Available online at http://vmls-book.stanford.edu/vmls.pdf.
2. Linear Algebra and Learning from Data, Gilbert Strang, Cambridge Uni. Press, 1st Edition, 2019.
3. IBM AI Engineering Professional Certificate (https://www.coursera.org/professional-certificates/ai-engineer).
4. Deep Learning with Python, François Chollet, Manning Publications, 2nd Edition, 2021.
5. Neural Networks and Deep Learning, Michael A. Nielsen, Determination Press, 2015. Available online at http://neuralnetworksanddeeplearning.com/
6. Matrix Methods: Applied Linear Algebra, Richard Bronson and Gabriel B. Costa, Academic Press, 3rd Edition, 2008.
7. Introduction to computational thinking (https://computationalthinking.mit.edu/Fall24/).

AME 5102: Applied Probability and Statistics

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Apply uncertainty in AI problems using probability theory.

CO2: Apply random variables for modelling uncertainty in AI systems.

CO3: Design and evaluate AI system decisions using estimation, hypothesis testing, and A/B testing.

Unit I **12 Hrs**

Probabilistic Thinking for AI Systems: Counting principles: multiplication rule; permutations and combinations; sampling with/without replacement; ordered vs unordered sampling. Binomial &

multinomial coefficients; Set theory for modelling: sample space; outcomes; events; Frequency-based definition of probability: equally likely vs non-equally likely outcomes; axioms of probability; Conditional probability: dependence vs independence; law of total probability; Bayes' rule (intuition driven).

AI Error Metrics: sensitivity; specificity; precision; ROC curve; precision-recall curve.

Unit II **12 Hrs**

Random Variables for AI Modelling: Discrete random variables: Bernoulli, geometric, binomial, negative binomial, and Poisson; PMF & CDF; Expectation and Variance.

Continuous random variables: uniform, normal, log normal, and exponential; PDF & CDF; Expectation and Variance.

Unit III **12 Hrs**

Sampling & Statistical Estimation in AI: Population vs sample; Statistic & sampling distribution; Sample mean and variance; Central Limit Theorem (intuition & relevance).

Estimation: point estimation; standard error; confidence interval interpretation.

Hypothesis testing: p-values; significance level; one-sample mean test; two-sample mean test; chi-squared test; A/B Testing in AI-Driven Systems. Likelihood and log-likelihood; maximum likelihood estimation; relationship between negative log-likelihood, KL divergence, and cross-entropy loss.

References

1. P. Nahin, *Digital Dice: Computational Solutions to Practical Probability Problems*. Princeton, NJ, USA: Princeton University Press.
2. C. M. Grinstead and J. L. Snell, *Introduction to Probability*, 2nd rev. ed. Providence, RI, USA: American Mathematical Society, 1997. [Online]. Available: https://chance.dartmouth.edu/teaching_aids/books_articles/probability_book/amsbook.mac.pdf
3. S. Ross, *A First Course in Probability*, 9th ed. Noida, India: Pearson Education India, 2013.
4. D. Rowntree, *Statistics without Tears: An Introduction for Non-Mathematicians*. London, U.K.: Penguin UK.
5. University of Florida, "Biostatistics Open Learning Textbook," [Online]. Available: https://bolt.mph.ufl.edu/6050-6052/
6. L. Wasserman, *All of Statistics: A Concise Course in Statistical Inference*. New York, NY, USA: Springer.

BDE 5101: Algorithms and Data Structures for Big Data

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Implement linked lists, stack, queues and binary search trees.

CO2: Implement dictionary, hash tables, graphs, sorting and searching.

CO3: Analyse the performance of algorithms and implement string and text processing techniques.

Unit I **10 hrs**

Elementary data structures: Implementation of lists, stacks, queues, binary search trees.

Unit II **14 hrs**

Sorting and Searching, Hash Tables, Graph: Quick sort, heap sort, merge sort, Linear search and binary search, Hashing and Dictionaries, Representation of graphs, Depth First Searching, Breadth First Searching, Minimum cost spanning tree, Single source shortest paths and all-pairs shortest path.

Unit III **12 hrs**

Algorithm specification and analysis techniques: Analysis of recursive programs, Solving recurrence equations, General solution for a large class of recurrence, String and text processing techniques, Data stream algorithms, Pattern-Matching Algorithms, Text Compression, Tries Sampling, Random Projections, Basic Algorithmic Techniques, Group Testing, Tree Method and Graph sketching.

References

1. Introduction to Algorithms - Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest. MIT Press.
2. Data Structures and Algorithms - Aho, Hopcroft and Ulmann. Pearson Publishers.
3. Data Structures and Algorithms in Python - Michael T. Goodrich, Roberto Tamassia, and Michael H. Goldwasser. John Wiley & Sons.
4. Data Streams: Algorithms and Applications - S. Muthukrishnan. Foundations and Trends in Theoretical Computer Science archive, Volume 1 Issue 2, August 2005, Pages 117 – 236.
5. MOOCS: https://www.coursera.org/learn/algorithms-on-strings

AME 5052: Applied Machine Learning Lab

(L-T-P-C: 0-0-3-1)

Total 36 hours

Course Outcome

CO1: Apply different types of supervised machine learning algorithms to practical problems.

CO2: Apply different types of unsupervised machine learning algorithms to practical problems.

CO3: Analyse different types of machine learning paradigms.

Unit I **12 hrs**

Introduction to Machine Learning, Decision Trees, Linear Models: Basics of Python for data handling: Introduction to Python libraries for machine learning (NumPy, Pandas, Matplotlib, Scikit-learn). Understanding datasets and data preprocessing techniques such as handling missing values, feature scaling, and encoding categorical variables.

Decision tree model of learning: Implementation of classification and regression using decision trees. Application of splitting criteria such as entropy, information gain, and Gini impurity. Analysis of overfitting and implementation of pruning techniques.

Unit II **12 hrs**

Linear regression: Implement linear regression model, apply different metrics (MSE, RMSE, R-sqaure).

Logistic regression for binary classification: Implement logistic regression model, apply sigmoid function, decision boundary. Apply regularization methods (Ridge and Lasso).

Nearest neighbours' algorithms: Apply K-NNN algorithms for the practical problems.

Probabilistic modelling: Implement maximum likelihood estimation (MLE) of parameters, Naive Bayes model for classification.

Model interpretation, evaluation and validation: Performance metrics such as accuracy, precision, recall, F1-score, ROC curve and AUC. Cross-validation techniques.

Apply bias/variance trade-off.

Visualize, compare, and contrast different models for practical problems.

Unit III **12 hrs**

Clustering techniques: Computing distances and similarities, implementation and analysis of K-means clustering and Hierarchical clustering, dendrogram construction and linkage methods.

Ensemble learning methods: Through coding, understand how ensemble methods in machine learning work, implementation and analysis of bagging techniques such as Random Forest and boosting algorithms.

Probabilistic clustering: Implementation of Gaussian Mixture Models (GMM) for clustering tasks.

Dimensionality reduction: Implementation and visualization using Principal Component Analysis (PCA).

Visualize, compare, and contrast different models for practical problems.

References

1. G. James, D. Witten, T. Hastie, R. Tibshirani, and J. Taylor, An Introduction to Statistical Learning: with Applications in Python, Springer Texts in Statistics. Cham, Switzerland: Springer, 2023. Online resource available at https://www.statlearning.com/

2. L. G. Serrano, Grokking Machine Learning, S. Thrun, Foreword, Manning Publications, 2021. Online resource from Manning Publications available at https://www.manning.com/books/grokking-machine-learning
3. J. VanderPlas, Python Data Science Handbook, Sebastopol, CA: O'Reilly Media, 2016.
4. A Course in Machine Learning, Hal Daumé III – Online resource available at http://ciml.info/
5. Module: Introduction to Machine Learning (https://www.intel.com/content/www/us/en/developer/tools/oneapi/training/academic-program/educators/intro-machine-learning-training-kit.html).
6. Learning path: Understand data science for machine learning (https://learn.microsoft.com/en-us/training/paths/understand-machine-learning/).

AME 5053: Principles of Natural Language Processing Lab

(L-T-P-C: 0-0-3-1)

Total 36 hours

Course Outcome

CO1: Implement text processing techniques for information extraction & retrieval.

CO2: Apply state-of-the-art word vectorization models for real-world problems.

CO3: Formulate and apply deep learning-based approaches for natural language processing tasks.

Unit I **9 hrs**

Introduction to Text Processing: Implement information extraction using regular expressions, implement text processing tasks such as tokenization, normalization, stemming, and stop words-removal, implement information retrieval using term document incidence matrices, implement inverted index construction and query processing.

Unit II **12 hrs**

Language Modelling, Sentiment Analysis, Word Embeddings: Implement n-grams and n-gram probabilities, implement sentiment Analysis using Naïve Bayes, implement cosine similarity and understand its significance in NLP, implement word vectorization models such as TF-IDF, Word2Vec, & GloVe, and compare them for real-world applications, Visualize word embeddings.

Unit III **15 hrs**

NLP with Deep Learning, Applications of Natural Language Processing: Implement sequence models for NLP tasks using Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks. Implement attention-based sequence-to-sequence models for text processing tasks. Implement transformer-based language models such as encoder and decoder transformers.

Apply pre-trained transformer models such as BERT and perform fine-tuning for NLP applications such as text classification, question answering, or text summarization.

References

1. IBM, “AI Engineering Professional Certificate.” [Online]. Available: https://www.coursera.org/professional-certificates/ai-engineer
2. S. Vajjala, B. Majumder, A. Gupta, and H. Surana, *Practical Natural Language Processing*. Sebastopol, CA, USA: O’Reilly Media, Jun. 2020.
3. D. Jurafsky and J. H. Martin, *Speech and Language Processing*, 3rd ed. (draft). [Online]. Available: https://web.stanford.edu/~jurafsky/slp3/
4. S. Bird, E. Klein, and E. Loper, *Natural Language Processing with Python: Analyzing Text with the Natural Language Toolkit*, 1st ed. London, U.K.: ISTE Ltd., 2017. [Online]. Available: https://www.nltk.org/book/
5. Y. Goldberg, *A Primer on Neural Network Models for Natural Language Processing*. [Online]. Available: http://faculty.cse.tamu.edu/huangrh/Spring18/nnlp.pdf
6. D. Rao and B. McMahan, *Natural Language Processing with PyTorch*, 1st ed. Sebastopol, CA, USA: O’Reilly Media, 2019.

AME 5151: Applied Linear Algebra Lab
(L-T-P-C: 0-0-3-1)
Total 36 hours

Course Outcome

CO1: Apply Python’s legacy libraries for coding vector- and matrix-related operations.

CO2: Implement models for real-life applications using linear algebra principles and analyze the results from a practical perspective.

CO3: Implement building blocks of deep learning models efficiently using linear algebra principles.

Unit I **9 hrs**

Vector Operations in Practice: Visualize vectors and vector-operations and relate them to their geometric descriptions; Implement and interpret results of distance-based clustering; Understand the importance of feature scaling; Compare words using embeddings: vector similarity computation; cosine similarity.

Unit II **15 hrs**

Matrix Computation Workflows: Understand how to perform matrix operations using Python; Implement and interpret matrix-operations using block representations; implement application examples from deep learning: data and weight matrices; forward propagation using matrix-based approach; batch forward propagation; attention mechanism; Implement and interpret dimension reduction using PCA; Solve systems of linear equations arising from application problems.

Unit III **12 hrs**

Optimization in Practice: Gradient descent implementation; Matrix-based optimization; Implement application examples from deep learning - backward propagation using matrix-based approach for binary and multiclass classification and regression

References

1. S. Boyd and L. Vandenberghe, *Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares*, 1st ed. Cambridge, U.K.: Cambridge University Press, 2018. [Online]. Available: http://vmls-book.stanford.edu/vmls.pdf
2. G. Strang, *Linear Algebra and Learning from Data*, 1st ed. Cambridge, U.K.: Cambridge University Press, 2019.
3. IBM, “AI Engineering Professional Certificate,” Coursera. [Online]. Available: https://www.coursera.org/professional-certificates/ai-engineer
4. F. Chollet, *Deep Learning with Python*, 2nd ed. Shelter Island, NY, USA: Manning Publications, 2021.
5. M. A. Nielsen, *Neural Networks and Deep Learning*. Determination Press, 2015. [Online]. Available: http://neuralnetworksanddeeplearning.com/
6. R. Bronson and G. B. Costa, *Matrix Methods: Applied Linear Algebra*, 3rd ed. Burlington, MA, USA: Academic Press, 2008.
7. Massachusetts Institute of Technology, “Introduction to Computational Thinking,” 2024. [Online]. Available: https://computationalthinking.mit.edu/Fall24/

AME 5152: Applied Probability and Statistics Lab

(L-T-P-C: 0-0-3-1)

Total 36 hours

Course Outcome

CO1: Analyse probabilistic simulation for AI & ML systems.

CO2: Apply discrete and continuous random variables to model real-world uncertainty.

CO3: Apply statistical inference for AI system evaluation.

Unit I **12 Hrs**

Data Sampling & Probabilistic Simulation: Simulation of real-world random experiments; Sampling with and without replacement; Verification of: multiplication rule, conditional probability, law of total probability, and Bayes’ rule.

Error metrics: sensitivity, specificity, and precision; Construction and interpretation of confusion matrix; ROC & precision-recall curve generation and interpretation under class imbalance.

Implement information-theoretic measures for: feature selection; decision tree splitting intuition; measuring uncertainty in class distributions.

Unit II **12 Hrs**

Random Variable Modelling: Simulation and visualization of discrete random variables: generate samples; plot empirical vs theoretical PMF; compute expectation and variance empirically and analytically.

Simulation and visualization of continuous random variables: estimate mean and variance from samples; compare histogram with theoretical PDF; visualize effect of parameter changes.

Unit III **12 Hrs**

Sampling & A/B Testing Experiments: Sampling & Estimation: compute sample mean; sample variance; standard error; Construct confidence intervals: interpret interval meaning through simulation.

Hypothesis Testing: one-sample mean test; two-sample mean test; chi-squared test; p-value computation and interpretation; Type I and Type II error simulation and interpretation.

A/B Testing: simulate conversion-rate experiments; test statistical significance; evaluate business decision thresholds.

Computationally show the relationship between negative log-likelihood, KL divergence, and cross-entropy loss.

References

1. P. Nahin, *Digital Dice: Computational Solutions to Practical Probability Problems*. Princeton, NJ, USA: Princeton University Press.
2. C. M. Grinstead and J. L. Snell, *Introduction to Probability*, 2nd rev. ed. Providence, RI, USA: American Mathematical Society, 1997. [Online]. Available: https://chance.dartmouth.edu/teaching_aids/books_articles/probability_book/amsbook.mac.pdf
3. S. Ross, *A First Course in Probability*, 9th ed. Noida, India: Pearson Education India, 2013.
4. D. Rowntree, *Statistics without Tears: An Introduction for Non-Mathematicians*. London, U.K.: Penguin UK.
5. University of Florida, “Biostatistics Open Learning Textbook,” [Online]. Available: https://bolt.mph.ufl.edu/6050-6052/
6. L. Wasserman, *All of Statistics: A Concise Course in Statistical Inference*. New York, NY, USA: Springer.

BDE 5151: Algorithms and Data structures for Big Data Lab

(L-T-P-C: 0-0-3-1)

Total 36 hours

Course Outcome

CO1: Design programs for the implementation of linked lists, stack and queues.

CO2: Design programs for implementation of binary search tree, sorting and searching, dictionary and Hash Table.

CO3: Design programs for graphs and shortest path techniques.

Unit I **15 hrs**

Single linked and double linked list: Array based and linked list-based Stack. Array based and linked list-based Queue.

Unit II **12 hrs**

Binary tree and BST: Hash functions, open and closed Hash Tables.

Unit III **9 hrs**

Graph algorithms and shortest path.

References

1. Introduction to Algorithms - Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest. MIT Press.
2. Data Structures and Algorithms - Aho, Hopcroft and Ulmann. Pearson Publishers.
3. Data Structures and Algorithms in Python - Michael T. Goodrich, Roberto Tamassia, and Michael H. Goldwasser. John Wiley & Sons.
4. Data Streams: Algorithms and Applications - S. Muthukrishnan. Foundations and Trends in Theoretical Computer Science archive, Volume 1 Issue 2, August 2005, Pages 117 – 236.

MPT 5100: Mini Project - I
(L-T-P-C: 0-0-0-4)
Total 48 hours

Course Outcome

CO1: Identify and formulate a real-world, technologically or socially relevant problem.

CO2: Design, experiment and execute the mini project effectively as a team, using appropriate tools and methodologies.

CO3: Demonstrate the ability to communicate the findings through written reports and oral presentations.

Unit I **6 hrs**

Problem identification, literature survey, formation of detailed specifications.

Unit II **30 hrs**

Design and implementation of the proposed system.

Unit III **12 hrs**

Presentation and demonstration of the work carried out to a panel of experts.

References

1. Research Methodology, Methods and Techniques, C. R. Kothari, New Age International Publishers, 2nd Revised Edition, 2004. (https://www.jsscacs.edu.in/sites/default/files/Files/Statistics%20and%20Research%20by%20Kothari.pdf).
2. Project Management by Adrienne Watt - BCcampus Open Textbook project (https://www.opentextbooks.org.hk/system/files/export/15/15694/pdf/Project_Management_15694.pdf).
3. Research articles and Online Resources.

PSD 5100: Professional Skill Development - I
(L-T-P-C: 0-0-0-1)
Total 12 hours

Course Outcome

CO1: Assemble important themes in the field of engineering domain and present a topic for effective communication in oral and written forms.

CO2: Demonstrate self-leaning capabilities through certification in the domain of study.

Unit I **7 hrs**

Technical Seminar and Report Writing: Presenting a technical seminar in the classroom to an audience, where the speaker is assessed on the content delivered, conceptual understanding, and presentation skills such as audibility, eye contact, and memory. The report writing component includes identifying a topic of current relevance in the field of engineering, technology, or interdisciplinary domains. Students are expected to structure the report logically, write a clear abstract, develop well-organized content, and conclude with future scope. Proper citation of sources and preparation of a bibliography are essential aspects of the report.

Unit II **5 hrs**

Micro-Credential: Introduction to Data Science using R Module 1-4 from SWAYAM Plus (https://swayam-plus.swayam2.ac.in/courses/course-details?id=F_IITMPR_01).

(or)

Course-7: Generative AI and LLMs: Architecture and Data Preparation and Course-8: Gen AI Foundational Models for NLP & Language Understanding from the IBM AI Engineering Professional Certificate (https://www.coursera.org/professional-certificates/ai-engineer).

(or)

AWS Academy Generative AI Foundations [118816] – Educator: AWS Academy Generative AI Foundations is intended for students who seek an introduction to artificial intelligence with a particular emphasis on generative AI technologies. It provides fundamental concepts, capabilities, principles, use cases and associated AWS services and tools of generative AI in the context of the AWS cloud. (https://awsacademy.instructure.com//courses/118816).

(or)

R Programming for Data Analytics from SWAYAM Plus (https://swayam-plus.swayam2.ac.in/courses/course-details?id=F_360DIG_08).

(or)

Data Engineer Modules 1-5 from SWAYAM Plus (https://swayam-plus.swayam2.ac.in/courses/course-details?id=F_360DIG_08).

(or)

GitHub Foundations:

https://learn.microsoft.com/en-us/collections/o1njfe825p602p

(or)

Google IT Automation with Python Professional Certificate:

https://www.coursera.org/professional-certificates/google-it-automation#courses

(or)

Linux Fundamentals:

https://www.coursera.org/learn/linux-fundamentals

(or)

AWS Academy Cloud Foundations [110120]:

https://awsacademy.instructure.com//courses/110120

(or)

Python fundamentals

https://swayam-plus.swayam2.ac.in/courses/course-details?id=F_360DIGI_08

(or)

Interpersonal Skills - advanced

https://swayam-plus.swayam2.ac.in/courses/course-details?id=F_WADHW_23

Reference

1. Research articles and Online Resources.

Semester II

AME 5202: Deep Learning

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Design and analyse deep learning models for application problems.

CO2: Formulate techniques for improving the way neural networks learn.

CO3: Select appropriate deep learning models for application problems.

Unit I **12 hrs**

One hidden layer neural network: Architecture and notation - The role of activation functions and their gradients – Loss functions: binary cross entropy loss, categorical cross entropy loss, focal loss, dice loss, and weighted loss functions – Deep L-layer neural network: architecture, notation, and building blocks - The importance of deep representations – The learning slowdown problem in deep neural networks: backward propagating gradient shrinkage due to softmax and activation function gradients, vanishing and exploding gradients - Improving the way deep neural networks learn: weight initialization, L1/L2 & dropout regularizations, residual & skip connections, batch normalization, adaptive learning, learning rate scheduling, gradient clipping, early stopping, and data augmentation.

Unit II **12 hrs**

Word vectors and language models: Word embeddings – Transformers: single and multi-head attention mechanisms, language models, vision transformers, generative image transformers, and multimodal transformers.

Unit III **12 hrs**

Advanced Topics in Deep Learning: Convolutional neural networks - Transfer Learning - Generative adversarial networks – Autoencoders – Knowledge Distillation - Quantization.

References

1. Neural Networks and Deep Learning, Michael Nielsen – Determination Press – Available online at http://neuralnetworksanddeeplearning.com/index.html
2. Deep Learning with Python, François Chollet, Manning Publications, 2nd Edition, 2021.
3. Deep Learning Foundations and Concepts, Christopher M. Bishop and Hugh Bishop, Springer, 2024.
4. IBM AI Engineering Professional Certificate (https://www.coursera.org/professional-certificates/ai-engineer).

5. Lecture slides of Prof. Andrew Ng – Stanford University – Available online at https://cs230.stanford.edu/syllabus/
6. Tensor Flow for Deep Learning Paperback, Reza Zadeh & Bharath Ramsundar, O'Reilly, 2018.
7. Deep Learning, Ian Goodfellow, Yoshua Bengio, and Aaron Courville – MIT Press – Available online at http://www.deeplearningbook.org/
8. Learning path: Microsoft Azure AI Fundamentals: Explore natural language processing (https://learn.microsoft.com/en-us/training/paths/explore-natural-language-processing/).

AME 5204: Applied Computer Vision
(L-T-P-C: 3-0-0-3)
Total 36 hours

Course Outcome

CO1: Understand visual feature representations in deep learning.

CO2: Analyse deep vision models for classification and detection.

CO3: Apply vision models in real-time deployment pipelines.

Unit I **10 hrs**

Foundations of Visual Representation: Image representation, Basic building block of CNN, CNN Architecture, convolution-based feature extraction, deep visual embeddings, transfer learning concepts, dataset efficiency, and comparison between CNNs and vision transformers.

Unit II **14 hrs**

Deep Vision Models: CNN architectures, region-based detection models, vision transformers, multimodal vision models, feature map learning, skip connections, loss functions, and fine-tuning pretrained visual models.

Unit III **12 hrs**

Vision Systems in Practice: Image classification pipelines, object detection, segmentation workflows, multimodal image-text systems, latency-performance tradeoffs, edge inference, and deployment of real-time vision models.

References

1. Learning path: Microsoft Azure AI Fundamentals: Explore computer vision (https://learn.microsoft.com/en-in/training/paths/explore-computer-vision-microsoft-azure/).
2. Generative AI for Everyone from DeepLearning.AI (https://www.coursera.org/learn/generative- ai-for-everyone).
3. IBM AI Engineering Professional Certificate (https://www.coursera.org/professional-certificates/ai-engineer).

4. Lecture slides of Prof. Fei-Fei Li – Stanford University – Available online at http://cs231n.stanford.edu/
5. A Guide to Convolutional Neural Networks for Computer Vision, Salman Khan, Hossein Rahmani, Syed Afaq Ali Shah, and Mohammed Bennamoun, Morgan & Claypool Publishers, 2018.
6. Lecture slides of Prof. Fei-Fei Li – Stanford University – Available online at http://cs231n.stanford.edu/
7. Neural Networks and Deep Learning, Michael Nielsen, Determination Press – Available online at http://neuralnetworksanddeeplearning.com/index.html
8. Computer Vision: Algorithms and Applications, Richard Szeliski, Springer, 2011 – Online resource from Springer available at http://szeliski.org/Book/

AME 5205: Advanced Generative AI and Agentic Systems

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Design knowledge-augmented generative AI systems using retrieval and embedding models.

CO2: Evaluate advanced large language model architectures and parameter-efficient fine-tuning techniques.

CO3: Design agentic AI systems and assess reliability, safety, and ethical considerations in generative AI applications.

Unit I **10 Hrs**

Large Language Models (LLMs): Large Language Models (LLM), Prompt Engineering, Limitations of LLM, Knowledge-augmented generative AI approaches: Retrieval Augmented Generation (RAG) architectures and modular RAG pipeline design. Dense retrieval methods and embedding-based search. Document chunking strategies and semantic indexing.

Hallucination in generative AI: causes, detection techniques, and evaluation of faithfulness and groundedness.

Applications of retrieval-based generative AI systems.

Unit II **12 Hrs**

Advanced Large Language Model Architectures: Large Language Model architectures: GPT architecture and comparison of modern open and closed LLM architectures. Long context handling techniques, Instruction tuning and alignment methods including Reinforcement Learning from Human Feedback (RLHF), Reinforcement Learning from AI Feedback (RLAIF).

Parameter-efficient fine-tuning techniques including LoRA and QLoRA.

Unit III **14 Hrs**

Agentic AI and Autonomous Generative Systems: Design principles of agentic AI, Tree-of-Thought reasoning, Memory architectures in AI agents, Multimodal agent systems, Speech-enabled conversational agents.

Reliability and safety in agentic systems: Agent failure modes, Self-correction and verification loops, Safety guardrails

Responsible, Secure Generative AI Systems: Ethical and responsible generative AI, Bias and fairness in foundation models, Explainability challenges in generative AI, Responsible AI frameworks , Governance and regulatory considerations.

References

1. D. Foster, Generative Deep Learning: Teaching Machines to Paint, Write, Compose, and Play, 2nd ed. Sebastopol, CA, USA: O'Reilly Media, 2019.
2. C. M. Bishop and H. Bishop, Deep Learning: Foundations and Concepts. Cham, Switzerland: Springer, 2024.
3. J. Alammar and M. Grootendorst, Hands-On Large Language Models: Language Understanding and Generation. Sebastopol, CA, USA: O'Reilly Media, 2024.
4. Hugging Face, "Transformers Documentation." [Online]. Available:https://huggingface.co/docs/transformers
5. Hugging Face, "PEFT: Parameter Efficient Fine-Tuning." [Online]. Available: https://huggingface.co/docs/peft
6. DeepLearning.AI, "Generative AI for Everyone," Coursera. [Online]. Available: https://www.coursera.org/learn/generative-ai-for-everyone
7. TransformaTech Institute, Creating Production-Ready LLMs: Building, Optimizing, and Deploying Large Language Models, 2024.

Elective I (Theory)

AME 5232: Reinforcement Learning

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Distinguish reinforcement learning from non-interactive machine learning.

CO2: Formulate an application problem as a reinforcement learning problem.

CO3: Apply state of the art reinforcement learning algorithms.

Unit I **10 hrs**

The Reinforcement Learning Framework: Examples and elements of reinforcement learning - Limitations and scope of reinforcement learning - History of reinforcement learning. Finite Markov decision process: the agent-environment interface, goals and rewards, returns, state & action value functions, Bellman equations. Optimal value functions: Bellman optimality equation, model-based methods: value iteration and policy iteration.

Unit II **16 hrs**

Model-free Reinforcement Learning: Importance of exploration - Monte Carlo methods - Temporal difference methods: Sarsa and Q-learning – Value function methods: deep Q-learning.

Unit III **10 hrs**

Advanced Reinforcement Learning Techniques: Policy gradient methods - Actor-critic methods - Policy-based vs. value-based methods.

References

1. Mathematical Foundations of Reinforcement Learning, Shiyu Zhao, Springer, 2025.
2. Deep Reinforcement Learning Hands-On, Maxim Lapan, 2nd Edition, Packt, 2020.
3. Reinforcement Learning: An Introduction, Richard S. Sutton and Andrew G. Barto, MIT Press, 2nd Edition - Available online at https://web.stanford.edu/class/psych209/Readings/SuttonBartoIPRLBook2ndEd.pdf
4. Fundamentals of Reinforcement Learning (University of Alberta, https://www.coursera.org/learn/fundamentals-of-reinforcement-learning)

BDE 5001: Architecture of Big Data Systems

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Apply various techniques to examine different types of data and understand lambda architecture.

CO2: Apply different tools and frameworks of Hadoop eco-system.

CO3: Apply Spark engine to process real-time data, implement batch and streaming data using Hadoop and Spark tools.

Unit I **12 hrs**

Classifying Big Data Characteristics and Big Data processing - the Lambda architecture, Analysis type, Processing methodology, Data Types, Data sources, Different data storing and processing layers and architecture.

Unit II **9 hrs**

Batch layer, Serving layer and Speed layer: Choosing a storage solution for the batch layer: Distributed file systems, Vertical partitioning, MapReduce - a paradigm for Big Data computing, Performance metrics for the serving layer, Speed layer.

Unit III **15 hrs**

Spark: Alternatives to MapReduce: Spark Architecture, Spark Session, DataFrame, Transformations and Actions, Spark SQL, Resilient Distributed Datasets (RDDs), Stream Processing and Machine Learning using Spark, Advantages and challenges of stream processing, Stream Processing Design Points, Streaming APIs, Structured Stream Processing High level M-Lib concepts and M-Lib in Action.

References

1. Big Data: Principles and best practices of scalable real-time data systems - Nathan Marz and James Warren. Manning Publisher.
2. Hadoop: The Definitive Guide: Storage and Analysis at Internet Scale – Tom White, O’Reilly Publication 4th Edition.
3. Spark: The Definitive Guide: Big Data Processing Made Simple – Bill Chambers, Matei Zaharia, O’Reilly Publication 1st Edition.
4. https://learn.microsoft.com/en-us/training/modules/explore-core-data-concepts/
5. https://learn.microsoft.com/en-us/training/modules/explore-fundamentals-stream-processing/
6. MOOCS: https://www.coursera.org/learn/machine-learning-with-apache-spark

CCE 5001: DevOps for Cloud
(L-T-P-C: 3-0-0-3)
Total 36 hours

Course Outcome

CO1: Understand existing Software Methodologies and DevOps Life Cycle stages.
CO2: Analyse and implement software product development using DevOps methodologies and tools.
CO3: Apply configuration management and automation templates to provision infrastructure and configure software in DevOps lifecycle.

Unit I **16 hrs**

DevOps Introduction: Understanding Development, Development SDLC- WaterFall & Agile, Understanding Operations - Dev vs Ops, DevOps to the rescue, What is DevOps, DevOps SDLC, Continuous Delivery model, DevOps tools for DevOps SDLC, DevOps Roles & Responsibilities.

Linux: Linux Introduction, Principles & Linux distro, Booting, Command line utilities & Basic commands, Linux File system, Filters & I/O Redirections, Users & Group administration, File permissions & Ownerships, Sudo, Software Management, Useful tools- ssh, telnet, scp, rsync, disk utils, backups etc, Service & Process management, Shell Scripting, Systems and HW stats, Linux Containers (lxc), Dockers, Kubernetes and Micro services.

Networking fundamentals: Components of computer networks, Classification- LAN, WAN, Peer to Peer network, Server based, Switches, Routers, Network Architecture, Protocols, Port numbers, DNS, DHCP, IP Addresses, IP Addresses & Subnet Masks, IP Address Ranges, Subnetting, Private vs Public networks, High Availability, Firewalls & NACL, Web Application Architecture, Infrastructure, Network layout, Services & Components, - Architecture from a DevOps perspective.

Unit II **12 hrs**

Automation, Orchestration & Config Management: Version control system with Git: What is VCS & why it is needed, DevOps use cases, Setup your own repo with git, Manage your code base/source code with GIT & GITHUB. Continuous Integration with Jenkins: Introduction to continuous integration, Build & Release and relation with DevOps, Understanding development and developers, Why Continuous integration, Jenkins introduction and setup, Jenkins projects/jobs, Jenkins plugins. Jenkins administration: Users, Nodes/slaves, Managing plugins, Managing software versions, Introduction, Phases, Java builds, Build and Release job/project setup. Nexus: Intro & Setup, Software versioning & Hosted repository, Integration with Jenkins, Continuous

Integration job/project setup. Complete Jenkins project: Packaging Artifacts, Static code Analysis, Tomcat setup Staging & productions, Artifacts deployments to webservers from Jenkins, Build Pipeline, Jenkins not just CI

tool anymore, More DevOps use cases of Jenkins.

Unit III **8 hrs**

Ansible: Configuration Management & Automation, What is Ansible & its features, Ansible setup on local & cloud, Understanding Ansible architecture & Execution, Inventory, Ad hoc commands, Automating change Management with Ad Hoc commands, Playbook Introduction, Ansible configuration with ansible, cfg-Ansible documentation, Modules, modules & lots of modules, Writing playbook for webserver & DB server deployments, Tasks, Variables, Templates, Loops, Handlers, Conditions, Register, Debugging, Ansible Roles, Identify server roles, Roles structure, Creating, Managing and executing roles, Ansible Galaxy, - Exploring Roles from Galaxy, Download Galaxy roles and integrate with your code, Ansible Advanced Execution, Improving execution time, Limiting and selecting tasks, Troubleshooting and Testing.

References

1. E. Foster-Johnson, J. C. Welch, and M. Anderson, *Beginning Shell Scripting (Programmer to Programmer)*. Indianapolis, IN, USA: Wrox Publications, 2005.
2. R. K. Michael, *Mastering Unix Shell Scripting: Bash, Bourne, and Korn Shell Scripting for Programmers, System Administrators, and UNIX Gurus*, 2nd ed. Hoboken, NJ, USA: Wiley, 2008.

3. B. Harwani, *UNIX & Shell Programming*. New Delhi, India: Oxford University Press, 2013.
4. J. F. Smart, *Jenkins: The Definitive Guide*. Sebastopol, CA, USA: O'Reilly Media, 2011.
5. M. Soni, *Jenkins Essentials*. Birmingham, U.K.: Packt Publishing, 2015.
6. R. Leszko, *Continuous Delivery with Docker and Jenkins*, 2nd ed. Birmingham, U.K.: Packt Publishing, 2019.
7. V. Kantse, *Implementing DevOps on AWS*. Birmingham, U.K.: Packt Publishing, 2017.
8. R. Smith, *Docker Orchestration*. Birmingham, U.K.: Packt Publishing, 2017.
9. A. Berg, *Jenkins Continuous Integration Cookbook*. Birmingham, U.K.: Packt Publishing, 2012.
10. K. S. Senthil, *Practical LXC and LXD: Linux Containers for Virtualization and Orchestration*. New York, NY, USA: Apress, 2017.
11. K. Ivanov, *Containerization with LXC*, 1st ed. Birmingham, U.K.: Packt Publishing, 2017.
12. K. Matthias and S. Kane, *Docker: Up & Running: Shipping Reliable Containers in Production*, 1st ed. Sebastopol, CA, USA: O'Reilly Media, 2015.
13. "Introduction to DevOps," Udacity. [Online]. Available: https://www.my-mooc.com/en/mooc/intro-to-devops--ud611/

ESE 5234: Entrepreneurship

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Explain the importance of entrepreneurship and entrepreneurial development model, social responsibilities of business.

CO2: Describe Entrepreneurial Traits and Factors affecting Entrepreneurship process.

CO3: Discuss Business Start-up Process.

CO4: Summarize a business and marketing plan for entrepreneurs.

Unit I **6 hrs**

Introduction to Entrepreneurship: Meaning and Definition of Entrepreneurship-Employment vs. Entrepreneurship, Theories of Entrepreneurship, approach to entrepreneurship, Entrepreneurs vs. Manager.

Unit II **5 hrs**

Entrepreneurial Traits: Personality of an entrepreneur, Types of Entrepreneurs.

Unit III **6 hrs**

Process of Entrepreneurship: Factors affecting Entrepreneurship process.

Unit IV **7 hrs**

Business Start-up Process: Idea Generation, Scanning the Environment, Macro and Micro analysis.

Unit V **6 hrs**

Business Plan writing: Points to be considered, Model Business plan.

Unit VI **6 hrs**

Case studies: Indian and International Entrepreneurship.

References

1. NVR Naidu and T. Krishna Rao, "Management and Entrepreneurship", IK International Publishing House Pvt. Ltd 2008.
2. Mohanthy Sangram Keshari, "Fundamentals of Entrepreneurship", PHI Publications, 2005.

Elective II (Theory)

AME 5241: Applied Mathematics for Machine Learning

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Apply matrix decomposition techniques to practical problems.

CO2: Understand different methods for computing derivatives.

CO3: Apply state-of-the-art optimization algorithms used in machine learning libraries.

Unit I **14 hrs**

Matrix Decompositions and Applications: Matrix and tensor products - Determinant and trace - Eigen-decomposition and diagonalization - Cholesky decomposition - Singular value decomposition - Nonnegative matrix factorization.

Unit II **8 hrs**

Computing Derivatives: Differentiability - Symbolic differentiation - Finite differences - Automatic differentiation.

Unit III **14 hrs**

Continuous Optimization: Optimization using gradient descent - Constrained optimization and Lagrange multipliers - Convex optimization – Sub gradients - Stochastic gradient descent - Momentum methods.

References

1. Mathematics for Machine Learning by Marc Peter Deisenroth, Aldo Faisal, and Cheng Soon Ong – Online resource from Cambridge University Press available at https://mml-book.github.io/book/mml-book.pdf
2. Matrix Computations, Gene H. Golub and Charles F. Van Loan, Hindustan Book Agency, 4th Edition, 2015.
3. Deep Learning, Ian Goodfellow, Yoshua Bengio, and Aaron Courville, MIT Press,2017. – Available online at http://www.deeplearningbook.org/
4. Understanding Machine Learning: From Theory to Algorithms (UML), Shai Shalev-Shwartz and Shai Ben-David, Cambridge University Press, 1st Edition, 2014.

AME 5243: Advanced Applications of Probability and Statistics

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Apply linear and logistic regression models for practical problems and assess model performance.

CO2: Apply multivariate techniques for outlier detection, dimension reduction, and clustering, and interpret the results.

CO3: Apply time series modelling to solve real-world problems.

Unit I **12 hrs**

Linear and Logistic Regression: Simple linear regression – regression model, estimating and interpreting coefficients, accuracy of coefficient estimates and model, ANOVA, R2 statistic - Multiple linear regression – estimating coefficients, qualitative predictors, interaction effects, potential problems - Logistic regression – binary and multinomial logistic regression models, estimating and interpreting coefficients, assessing model calibration and discrimination, area under the ROC.

Unit II **15 hrs**

Multivariate Distributions: Mean vector, covariance and correlation matrices – population vs. sample - The multivariate Gaussian – joint-, marginal-, and conditional distributions, Mahalanobis distance and outliers - Properties of the multivariate Gaussian - Parameter estimation: maximum likelihood estimation (MLE) and maximum aposteriori estimation (MAP).

Principal Component Analysis, Cluster Analysis: Geometric intuition of principal components - Maximum variance perspective – algebraic setup, eigenvectors, and eigenvalues of sample correlation matrix - Interpretation and application of principal components for dimension reduction.

Dissimilarity measures for mixed data types - Partition around medoids (PAM) vs. K-means algorithms - Selecting the number of clusters.

Unit III **9 hrs**

Time Series Analysis: Time series concepts: resampling, smoothing, windowing, rolling average, trend, seasonality, stationarity, differencing, autocorrelation, partial autocorrelation – Autoregressive (AR), moving average (MA), autoregressive moving average (ARMA), and autoregressive integrated moving average (ARIMA) models - Validating time series predictions.

References

1. An Introduction to Statistical Learning with Applications in R, Gareth James, Daniela Witten, Trevor Hastie and Robert Tibshirani, Springer, 1st Edition, 2013, Corr. 7th printing 2017 Edition.
2. Applied Multivariate Statistical Modeling from Swayam (NPTEL, https://onlinecourses.nptel.ac.in/noc23_ma26/preview).
3. An Introduction to Applied Multivariate Analysis with R, Brian Everitt and Torsten Hothorn-Springer Publications,1st Edition, 2011.
4. Machine Learning - A Probabilistic Perspective, Kevin P. Murphy, The MIT Press, 1st Edition, 2012.
5. Mathematics for Machine Learning, Marc Peter Deisenroth, Aldo Faisal, and Cheng Soon Ong, Cambridge University Press, 2020. – Online resource from Cambridge University Press available at https://mml-book.github.io/book/mml-book.pdf

AME 5244: AI/ML on Cloud
(L-T-P-C: 3-0-0-3)
Total 36 hours

Course Outcome

CO1: Apply AI and ML techniques on cloud platforms to enable data-driven decision-making within organisations.

CO2: Analyse MLOps practices, cloud infrastructure components, and deployment architectures for AI/ML systems.

CO3: Create strategies for managing, monitoring, and scaling ML workflows in cloud environments that align with business objectives.

Unit I **10 hrs**

Cloud Computing Foundations for Machine Learning: Introduction to Machine Learning on Cloud, Cloud Computing Fundamentals, Comparative Analysis of Cloud Platforms for AI/ML, Cloud Data Architecture, Data Engineering and Data Pipeline for ML.

Unit II **12 hrs**

Cloud Infrastructure and MLOps Foundations: Containerization for ML Systems, Container Orchestration, Microservices Architecture for ML, MLOps Fundamentals, Continuous Integration /

Continuous Deployment/ Continuous Training (CI/CD/CT), Feature Stores and Model Training Infrastructure

Unit III **14 hrs**

Production ML Systems: ML Model Deployment, Experiment Tracking and Model Registry, Cost Optimization, Emerging Trends in Cloud ML, End-to-End Cloud ML Architecture, Monitoring and Scaling ML Systems, Security and Governance

References

1. "Designing Machine Learning Systems" by Chip Huyen (O'Reilly, 2022)
2. "Machine Learning Engineering"Andriy Burkov (2020) https://leanpub.com/MLE (Free)
3. AWS Well-Architected Framework (ML Lens) https://docs.aws.amazon.com/wellarchitected/latest/machine-learning-lens/ (Free)
4. Building Machine Learning Pipelines – Hannes Hapke & Catherine Nelson (O'Reilly, 2020)
5. "Fundamentals of Data Engineering" by Reis & Housley (2022)
6. "Practical MLOps" by Noah Gift & Alfredo Deza (O'Reilly, 2021)

ESE 5244: IT Project Management

(L-T-P-C: 3-0-0-3)

Total 36 hours

Course Outcome

CO1: Illustrate the importance of project planning.

CO2: Discuss and demonstrate various tools applicable for different phases of the software project.

CO3: Illustrate the importance of Change management.

CO4: Illustrate the importance of teamwork and demonstrate the usefulness of tools in effective handling of the project.

CO5: Compare the problems in projects that don't follow the process and discuss new trends in software life cycle.

Unit I **3 hrs**

Software Project Planning: Understand the Project Needs, Create the Project Plan, Diagnosing Project Planning Problems.

Unit II **3 hrs**

Estimation: Elements of a Successful Estimate, Wideband Delphi Estimation, Other Estimation Techniques, Diagnosing Estimation Problems.

Unit III **3 hrs**

Project Schedules: Building the Project Schedule, Managing Multiple Projects, Use the Schedule to Manage Commitments, Diagnosing Scheduling Problems.

Unit IV **4 hrs**

Reviews: Inspections, Deskchecks, Walkthroughs, Code Reviews, Pair Programming, Use Inspections to Manage Commitments, Diagnosing Review Problems.

Unit V **4 hrs**

Software Requirements: Requirements Elicitation, Use Cases, Software Requirements Specification, Change Control, Introduce Software Requirements Carefully, Diagnosing Software Requirements Problems.

Unit VI **4 hrs**

Design and Programming: Review the Design, Version Control with Subversion, Refactoring, Unit Testing, Use Automation, Be Careful with Existing Projects, Diagnosing Design and Programming Problems.

Unit VII **4 hrs**

Software Testing: Test Plans and Test Cases, Test Execution, Defect Tracking and Triage, Test Environment and Performance Testing, Smoke Tests, Test Automation, Postmortem Reports, Using Software Testing Effectively, Diagnosing Software Testing Problems.

Unit VIII **3 hrs**

Understanding Change: Why Change Fails, How to Make Change Succeed.

Unit IX **2 hrs**

Management and Leadership: Take Responsibility, Do Everything Out in the Open, Manage the Organization, Manage Your Team.

Unit X **3 hrs**

Managing an Outsourced Project: Prevent Major Sources of Project Failure, Management Issues in Outsourced Projects, Collaborate with the Vendor.

Unit XI **3 hrs**

Process Improvement: Life Without a Software Process, Software Process Improvement, Moving Forward.

References

1. "Applied Software Project Management", Jennifer Greene, Andrew Stellman (O'Reilly Publications), 2005.
2. "The Art of Project Management", Scott Berkun (O'Reilly Publications), 2005.

AME 5252: Deep Learning Lab
(L-T-P-C: 0-0-3-1)
Total 36 hours

Course Outcome

CO1: Apply state-of-the-art deep learning tools and libraries.

CO2: Implement and investigate techniques for improving the way neural networks learn.

CO3: Apply and analyze deep learning models for real-world problems.

Unit I **12 hrs**

Deep Neural Networks: Visualize different nonlinear activation functions, Implement forward and backward propagation for a shallow neural network efficiently using a matrix-based approach, Implement gradient descent method for a shallow neural network, Implement different loss functions and study their effectiveness for real-world problems, Numerically investigate the learning slowdown problem in deep neural networks: backward propagating gradient shrinkage due to softmax and activation function gradients, vanishing and exploding gradients, Numerically implement and investigate the effect of weight initialization, L1/L2 & dropout regularizations, residual & skip connections, batch normalization, adaptive learning, learning rate scheduling, gradient clipping, early stopping, and data augmentation.

Unit II **12 hrs**

Transformer Networks: Implement and understand word embeddings from language models, Implement and understand single and multi-head attention mechanisms, Implement different types of transformers such as vision transformers, generative image transformers, and multimodal transformers for real-world problems.

Unit III **12 hrs**

Advanced Topics in Deep Learning: Implement and understand transfer Learning from existing state-of-the-art model, Implement Generative adversarial networks and autoencoders for real-world problems such as anomaly detection, develop skills in building and implementing problem-specific small-scale models using knowledge distillation and quantization.

References

1. Neural Networks and Deep Learning, Michael Nielsen – Determination Press – Available online at http://neuralnetworksanddeeplearning.com/index.html

2. Deep Learning with Python, François Chollet, Manning Publications, 2nd Edition, 2021.
3. Deep Learning Foundations and Concepts, Christopher M. Bishop and Hugh Bishop, Springer, 2024.
4. IBM AI Engineering Professional Certificate (https://www.coursera.org/professional-certificates/ai-engineer).
5. Lecture slides of Prof. Andrew Ng – Stanford University – Available online at https://cs230.stanford.edu/syllabus/
6. Tensor Flow for Deep Learning Paperback, Reza Zadeh & Bharath Ramsundar, O'Reilly, 2018.
7. Deep Learning, Ian Goodfellow, Yoshua Bengio, and Aaron Courville – MIT Press – Available online at http://www.deeplearningbook.org/
8. Learning path: Microsoft Azure AI Fundamentals: Explore natural language processing (https://learn.microsoft.com/en-us/training/paths/explore-natural-language-processing/).

AME 5254: Applied Computer Vision Lab
(L-T-P-C: 0-0-3-1)
Total 36 Hours

Course Outcome

CO1: Implement transfer learning for vision tasks
CO2: Apply detection and segmentation workflows
CO3: Implement vision models for real-time inference

Unit I **10 hours**

Transfer Learning for Vision: Implementation of transfer learning for image classification using pretrained CNN models including data augmentation and model evaluation.

Unit II **14 hours**

Detection & Segmentation: Object detection and segmentation workflows using deep vision models with performance benchmarking.

Unit III **12 hours**

Vision Deployment: Real-time inference, latency benchmarking, edge deployment, API-based deployment, and monitoring of vision model performance.

References

1. Learning path: Microsoft Azure AI Fundamentals: Explore computer vision (https://learn.microsoft.com/en-in/training/paths/explore-computer-vision-microsoft-azure/).
2. Generative AI for Everyone from DeepLearning.AI (https://www.coursera.org/learn/generative- ai-for-everyone).

3. IBM AI Engineering Professional Certificate (https://www.coursera.org/professional-certificates/ai-engineer).
4. Lecture slides of Prof. Fei-Fei Li – Stanford University – Available online at http://cs231n.stanford.edu/
5. A Guide to Convolutional Neural Networks for Computer Vision, Salman Khan, Hossein Rahmani, Syed Afaq Ali Shah, and Mohammed Bennamoun, Morgan & Claypool Publishers, 2018.
6. Lecture slides of Prof. Fei-Fei Li – Stanford University – Available online at http://cs231n.stanford.edu/
7. Neural Networks and Deep Learning, Michael Nielsen, Determination Press – Available online at http://neuralnetworksanddeeplearning.com/index.html
8. Computer Vision: Algorithms and Applications, Richard Szeliski, Springer, 2011 – Online resource from Springer available at http://szeliski.org/Book/

AME 5255: Advanced Generative AI and Agentic Systems Lab

(L-T-P-C: 0-0-3-1)

Total 36 Hours

Course Outcome

CO1: Implement retrieval-augmented generative AI systems using embedding models and vector databases.

CO2: Apply prompt engineering and parameter-efficient fine-tuning techniques to adapt large language models.

CO3: Design simple agentic AI workflows and evaluate reliability, safety, and ethical aspects of generative AI systems.

Unit I **10 hrs**

Introduction to generative AI tools: prompt engineering experiments, exploring limitations and hallucinations in large language models, generating text using models from Hugging Face Transformers, creating text embeddings using SentenceTransformers, semantic similarity search using vector databases such as FAISS, document chunking and indexing strategies, implementation of a basic Retrieval-Augmented Generation (RAG) pipeline using LangChain, evaluation of RAG outputs for faithfulness and hallucination reduction.

Unit II **12 hrs**

Generative Learning Algorithms: Exploration of large language model architectures, instruction-based prompting techniques, implementing prompt templates and prompt chaining, demonstration of instruction tuning workflows, parameter-efficient fine-tuning using LoRA, fine-tuning using the PEFT library, comparison of base model responses versus adapted model responses, evaluating model outputs for accuracy and relevance.

Unit III **14 hrs**

Design of simple AI agents using LangChain: implementation of multi-step reasoning using Tree-of-Thought prompting, building conversational agents with memory modules, designing speech-enabled conversational agents using APIs, monitoring AI agent responses and failure modes, implementing guardrails and safety filters, evaluating bias and fairness in generated outputs, basic explainability and responsible AI considerations in generative systems.

References

1. L. Wu et al., *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*, arXiv, 2020.
2. C. Huyen, *AI Engineering: Building Applications with Foundation Models*. Sebastopol, CA, USA: O’Reilly Media, 2024.
3. S. Raschka, *Build a Large Language Model from Scratch*. Birmingham, U.K.: Packt Publishing, 2024.
4. T. Tunstall, L. von Werra and T. Wolf, *Natural Language Processing with Transformers*. Sebastopol, CA, USA: O’Reilly Media, 2022.
5. Hugging Face, *Transformers Documentation*. [Online]. Available: https://huggingface.co/docs

Elective I (Lab)

AME 5282: Reinforcement Learning Lab

(L-T-P-C: 0-0-3-1)

Total 36 hours

Course Outcome

CO1: Implement real-life problems using Markov decision processes and solve them computationally.
CO2: Apply model free reinforcement learning techniques for real-world problems.
CO3: Apply advanced reinforcement learning algorithms.

Unit I **12 hrs**

The Reinforcement Learning Framework: Implement building blocks for solving a reinforcement learning task using the Gym library, Implement Markov decision process models, Implement iterative policy evaluation using Bellman equations and iterative optimal policy estimation using value iteration and policy iteration.

Unit II **15 hrs**

Model-free Reinforcement Learning: Implement Monte Carlo and temporal difference methods for solving reinforcement learning tasks, Apply Sarsa and deep Q-learning for real-world problems.

Unit III **9 hrs**

Advanced Reinforcement Learning Techniques: Apply advanced reinforcement learning techniques such as policy gradient methods, actor–critic methods, and policy-/value-based methods for real-world problems.

References

1. Mathematical Foundations of Reinforcement Learning, Shiyu Zhao, Springer, 2025.
2. Deep Reinforcement Learning Hands-On, Maxim Lapan, 2nd Edition, Packt, 2020.
3. Reinforcement Learning: An Introduction, Richard S. Sutton and Andrew G. Barto, MIT Press, 2nd Edition –Available online at https://web.stanford.edu/class/psych209/Readings/SuttonBartoIPRLBook2ndEd.pdf
4. Fundamentals of Reinforcement Learning (University of Alberta, https://www.coursera.org/learn/fundamentals-of-reinforcement-learning).

BDE 5051: Architecture of Big Data Systems Lab

(L-T-P-C: 0-0-3-1)

Total 36 hours

Course Outcome

CO1: Experiment with different tools and frameworks of Hadoop eco-system.

CO2: Experiment with Spark Engine to process real-time data.

CO3: Design applications to handle batch and streaming data using Hadoop and Spark tools

Unit I **15 hrs**

Using HDFS commands to get familiarize with HDFS environment: Transfer structured data to and from HDFS, Creating Hive tables in HDFS and running SQL statements, Developing and running MapReduce programs.

Unit II **12 hrs**

Creating RDDs in spark: Creating Spark DataFrames, loading, transforming and performing actions in Spark environment. Handling streaming data using structured streaming.

Unit III **9 hrs**

Developing applications using Hadoop tools: Develop applications using Spark environment.

References

1. Big Data: Principles and best practices of scalable real-time data systems - Nathan Marz and James Warren. Manning Publisher.
2. Hadoop: The Definitive Guide: Storage and Analysis at Internet Scale – Tom White, O'Reilly Publication 4th Edition.

3. Spark: The Definitive Guide: Big Data Processing Made Simple – Bill Chambers, Matei Zaharia, O'Reilly Publication 1st Edition.

CCE 5051: DevOps for Cloud Lab

(L-T-P-C: 0-0-3-1)

Total 36 hours

Course Outcome

CO1: Implement Linux and version control system commands in DevOps Life Cycle stages.

CO2: Design CICD pipeline to automate software product development lifecycle.

CO3: Design configuration management and automation templates to provision infrastructure and configure software in DevOps lifecycle.

Unit I **12 hrs**

Understand the lifecycle stages of DevOps: product life cycle methodology, Explore the basic Linux commands, system utilities and services, Understand the basic architecture of software version controls like GIT, SVN, Setup a public git repo for an application, explore the GIT commands to perform clone, fork, pull, push, commit etc.

Unit II **15 hrs**

Demonstrate a continuous integration pipeline: using Jenkins master and slave architecture for web applications, Explore containerizing web applications using Docker containers, Setup a Kuberenetes cluster to deploy containerised application with high availability and load balancing.

Unit III **9 hrs**

Create Ansible playbook: for configuring Linux and window host, Setup a continuous monitoring service for managing IT assets.

References

1. E. Foster-Johnson, J. C. Welch, and M. Anderson, *Beginning Shell Scripting (Programmer to Programmer)*. Indianapolis, IN, USA: Wrox Publications, 2005.
2. R. K. Michael, *Mastering Unix Shell Scripting: Bash, Bourne, and Korn Shell Scripting for Programmers, System Administrators, and UNIX Gurus*, 2nd ed. Hoboken, NJ, USA: Wiley, 2008.
3. B. Harwani, *UNIX & Shell Programming*. New Delhi, India: Oxford University Press, 2013.
4. J. F. Smart, *Jenkins: The Definitive Guide*. Sebastopol, CA, USA: O'Reilly Media, 2011.
5. M. Soni, *Jenkins Essentials*. Birmingham, U.K.: Packt Publishing, 2015.
6. R. Leszko, *Continuous Delivery with Docker and Jenkins*, 2nd ed. Birmingham, U.K.: Packt Publishing, 2019.
7. V. Kantse, *Implementing DevOps on AWS*. Birmingham, U.K.: Packt Publishing, 2017.

8. R. Smith, *Docker Orchestration*. Birmingham, U.K.: Packt Publishing, 2017.
9. A. Berg, *Jenkins Continuous Integration Cookbook*. Birmingham, U.K.: Packt Publishing, 2012.
10. K. S. Senthil, *Practical LXC and LXD: Linux Containers for Virtualization and Orchestration*. New York, NY, USA: Apress, 2017.
11. K. Ivanov, *Containerization with LXC*, 1st ed. Birmingham, U.K.: Packt Publishing, 2017.
12. K. Matthias and S. Kane, *Docker: Up & Running: Shipping Reliable Containers in Production*, 1st ed. Sebastopol, CA, USA: O'Reilly Media, 2015.

ESE 5284: Entrepreneurship Lab
(L-T-P-C: 0-0-3-1)
Total 36 hours

Course Outcome

CO1: Understand prominence of entrepreneurship.

CO2: Develop use cases for building a business.

CO3: Evaluation of factors influencing business venture.

Unit I **6 hrs**

Study of use cases for need and prominence of entrepreneurship, associated decision-making process.

Unit II **6 hrs**

Study a report by the National Knowledge Commission on the importance of entrepreneurship in economic development.

Unit III **9 hrs**

Develop use cases for identifying and evaluating opportunities, developing business plan, assessment of resources, project appraisal and feasibility plan.

Unit IV **9 hrs**

Creating and starting venture includes legal requirements, marketing strategies, financial plans and human resources management.

Unit V **6 hrs**

Design a Case studies of Indian and International Entrepreneurship.

References

1. Management and Entrepreneurship, NVR Naidu and T. Krishna Rao, IK International Publishing House Pvt. Ltd, 2008.
2. Fundamentals of Entrepreneurship, Mohanthy Sangram Keshari, PHI Learning Pvt. Ltd., 2005.

Elective II (Lab)

AME 5291: Applied Mathematics for Machine Learning Lab

(L-T-P-C: 0-0-3-1)

Total 36 hours

Course Outcome

CO1: Apply matrix decomposition techniques for practical problems.

CO2: Implement and compare different methods for computing derivatives.

CO3: Apply state-of-the-art optimization algorithms used in machine learning libraries.

Unit I **12 hrs**

Matrix Decompositions and Applications: Implement matrix decompositions using block matrix representations, Implement and compare exact and approximate decompositions, implement codes to understand the optimization-centric view to matrix factorization, Interpret the factors arising out of matrix factorizations for real-life problems.

Unit II **9 hrs**

Computing Derivatives: Visualize differentiability concepts in 3D, implement symbolic differentiation for computing derivatives exactly, implement finite difference methods for approximating derivatives, implement automatic differentiation and compare it with other approaches.

Unit III **15 hrs**

Continuous Optimization: Solve continuous optimization problems using state of the art libraries, Visualize constrained optimization problems and solutions in 3D, Implement and visualize solutions of gradient descent method and its extensions for continuous optimization, understand implementations of continuous optimization algorithms used in state-of-the-art libraries.

References

1. Mathematics for Machine Learning by Marc Peter Deisenroth, Aldo Faisal, and Cheng Soon Ong – Online resource from Cambridge University Press available at https://mml-book.github.io/book/mml-book.pdf
2. Matrix Computations, Gene H. Golub and Charles F. Van Loan, Hindustan Book Agency, 4th Edition, 2015.
3. Deep Learning, Ian Goodfellow, Yoshua Bengio, and Aaron Courville, MIT Press,2017. – Available online at http://www.deeplearningbook.org/
4. Understanding Machine Learning: From Theory to Algorithms (UML), Shai Shalev-Shwartz and Shai Ben-David, Cambridge University Press, 1st Edition, 2014.

AME 5293: Advanced Applications of Probability and Statistics Lab

(L-T-P-C: 0-0-3-1)

Total 36 hours

Course Outcome

CO1: Apply linear and logistic regression models for practical problems and assess model performance.

CO2: Apply multivariate techniques for outlier detection, dimension reduction, and clustering, and analyse the results.

CO3: Appy time series modelling to real-world problems.

Unit I **9 hrs**

Linear and Logistic Regression: Use in-built functions in R to build linear models for practical problems, compute different performance metrics to assess model performance, Interpret model coefficients and investigate the effect of input variables on output through sensitivity analysis, Use in-built functions in R to build logistic regression models for practical binary classification problems and assess model performance.

Unit II **15 hrs**

Multivariate Distributions: Compute descriptive statistics of multivariate data, perform exploratory data analysis of multivariate data, identify outliers in multivariate data, Visualise and understand the properties of multivariate Gaussian data.

Principal Component Analysis, Cluster Analysis: Visualise the geometric interpretation of principal component analysis (PCA), Use in-built functions in R to perform PCA on multivariate data, Compare and contrast PCA for variance maximization vs. clustering of multivariate data, Cluster multivariate data with mixed data types using in-built functions in R.

Unit III **12 hrs**

Time Series Analysis: Understand and apply in-built functions and libraries in R for time series modelling, apply time series modelling to practical problems, Interpret the results of times series model predictions.

References

1. An Introduction to Statistical Learning with Applications in R, Gareth James, Daniela Witten, Trevor Hastie and Robert Tibshirani, Springer, 1st Edition, 2013, Corr. 7th printing 2017 Edition.
2. Applied Multivariate Statistical Modeling from Swayam (NPTEL, https://onlinecourses.nptel.ac.in/noc23_ma26/preview).
3. An Introduction to Applied Multivariate Analysis with R, Brian Everitt and Torsten Hothorn-Springer Publications,1st Edition, 2011.

4. Machine Learning - A Probabilistic Perspective, Kevin P. Murphy, The MIT Press, 1st Edition, 2012.
5. Mathematics for Machine Learning, Marc Peter Deisenroth, A Aldo Faisal, and Cheng Soon Ong, Cambridge University Press, 2020. – Online resource from Cambridge University Press available at https://mml-book.github.io/book/mml-book.pdf

AME 5294: AI/ML on Cloud Lab

(L-T-P-C: 0-0-3-1)

Total 36 hours

Course Outcome

CO1: Implement text processing techniques for information extraction & retrieval.

CO2: Apply state-of-the-art word vectorization models for real-world problems.

CO3: Formulate and apply deep learning-based approaches for natural language processing tasks.

Unit I **9 hrs**

Introduction to cloud platforms for machine learning, creating cloud-based development environments, dataset ingestion and cloud storage, building data pipelines for ML workflows, preprocessing and feature preparation using pandas and NumPy, training basic machine learning models on cloud notebook environments, experimentation using services from Amazon Web Services, Google Cloud Platform, or Microsoft Azure.

Unit II **12 hrs**

Containerization of machine learning applications using Docker, introduction to container orchestration using Kubernetes, designing microservice-based ML applications, implementing MLOps pipelines for model training and evaluation, automating ML workflows using CI/CD pipelines, introduction to feature stores and model training infrastructure.

Unit III **15 hrs**

Deployment of machine learning models as web services, implementing REST APIs for ML inference, experiment tracking and model versioning using MLflow, monitoring model performance and detecting model drift, scaling machine learning services using cloud infrastructure, cost optimization strategies for ML workloads, basic security and access control mechanisms for ML systems.

References

1. C. Huyen, Designing Machine Learning Systems: An Iterative Process for Production-Ready Applications. Sebastopol, CA, USA: O’Reilly Media, 2022.
2. A.Burkov, Machine Learning Engineering. 2020. [Online]. Available: https://leanpub.com/MLE

3. H. Hapke and C. Nelson, Building Machine Learning Pipelines. Sebastopol, CA, USA: O'Reilly Media, 2020.
4. N. Gift and A. Deza, Practical MLOps. Sebastopol, CA, USA: O'Reilly Media, 2021.
5. J. Reis and M. Housley, Fundamentals of Data Engineering. Sebastopol, CA, USA: O'Reilly

ESE 5294: IT Project Management Lab

(L-T-P-C: 0-0-3-1)

Total 36 hours

Course Outcome

CO1: Identify the skills and techniques required for project life cycle.

CO2: Apply essential principles associated with project management and their application in business environment.

CO3: Compare commonly available project management tools.

Unit I **12 hrs**

Software project management and its need, evaluation techniques, Project cost estimation techniques.

Unit II **12 hrs**

Activity scheduling techniques, cost monitoring, contract management and its necessity risk analysis.

Unit III **12 hrs**

Software quality enhancement techniques, people management in software environment and project team structure.

References

1. "Applied Software Project Management", Jennifer Greene, Andrew Stellman (O'Reilly Publications), 2005.
2. "The Art of Project Management", Scott Berkun (O'Reilly Publications), 2005.

MPT 5200: Mini Project - II

(L-T-P-C: 0-0-0-4)

Total 48 hours

Course Outcome

CO1: Identify and formulate a real-world, technologically or socially relevant problem.

CO2: Design, experiment and execute the mini project effectively as a team, using appropriate tools and methodologies.

CO3: Demonstrate the ability to communicate the findings through written reports and oral presentations.

Unit I **6 hrs**

Problem identification, literature survey, formation of detailed specifications.

Unit II **30 hrs**

Design and implementation of the proposed system.

Unit III **12 hrs**

Presentation and demonstration of the work carried out to a panel of experts.

References

1. Research Methodology, Methods and Techniques, C. R. Kothari, New Age International Publishers, 2nd Revised Edition, 2004. (https://www.jsscacs.edu.in/sites/default/files/Files/Statistics%20and%20Research%20by%20Kothari.pdf).
2. Project Management by Adrienne Watt - BCcampus Open Textbook project (https://www.opentextbooks.org.hk/system/files/export/15/15694/pdf/Project_Management_15694.pdf).
3. Research articles and Online Resources.

PSD 5200: Professional Skill Development - II
(L-T-P-C: 0-0-0-1)
Total 12 hours

Course Outcome

CO1: Develop the skills needed for approaching technical and HR interviews.
CO2: Demonstrate the depth of knowledge in the chosen field of study.
CO3: Use mathematical, reasoning, and domain specific skills to solve objective questionnaires in time.

Unit I **4 hrs**

Peer interviews, mock interviews.

Unit II **4 hrs**

Conduction of domain specific knowledge test using micro-credentials as follows:
Course-9: Generative AI Language Modeling with Transformers and Course-12: Fundamentals of AI Agents Using RAG and LangChain from the IBM AI Engineering Professional Certificate (https://www.coursera.org/professional-certificates/ai-engineer)
(or)

Data Engineer Modules 6-9 from SWAYAM Plus (https://swayam-plus.swayam2.ac.in/courses/course-details?id=F_360DIG_08)

(or)

Introduction to Data Science using R Modules 5-7 from SWAYAM Plus (https://swayam-plus.swayam2.ac.in/courses/course-details?id=F_IITMPR_01)

Unit III **4 hrs**

Analytical and logical reasoning.

References

1. R S Aggarwal. Quantitative Aptitude for Competitive Examinations. S Chand, 2017.
2. McDowell, Gayle Laakmann. Cracking the coding interview: 189 programming questions and solutions. CareerCup, LLC, 2015.
3. Domain specific tools and online resources.

Semesters III & IV

AME 6098: Project Work

(L-T-P-C: 0-0-0-30)

Total 360 hours

Course Outcome

CO1: Undertake innovative industry/research-oriented projects and perform feasibility analysis for finding solutions.

CO2: Implement and test the proposed design using appropriate framework, programming language and tools.

CO3: Demonstrate an ability to present and defend project work carried out to a panel of experts.

Unit I **30 hrs**

Problem identification, literature survey, formation of detailed requirement specification document.

Unit II **300 hrs**

Design and implementation of the proposed modules with specific test cases.

Unit III **30 hrs**

Detailed report of the work carried out, present, and defend the project work carried out to a panel of experts.

Reference

1. Research articles, domain specific tools and online resources.

Program Outcome and Course Outcome Mapping

Sl. No.	Course Code	Course Name	Credits	PO1	PO2	PO3	PO4	PO5
1	AME 5002	Applied Machine Learning	3	*		*	*	
2	AME 5003	Principles of Natural Language Processing	3			*	*	
3	AME 5101	Applied Linear Algebra	3			*	*	
4	AME 5102	Applied Probability and Statistics	3	*		*		
5	BDE 5101	Algorithms and Data Structures for Big Data	3	*			*	
7	AME 5052	Applied Machine Learning Lab	1	*		*	*	
6	AME 5053	Principles of Natural Language Processing Lab	1			*		*
8	AME 5151	Applied Linear Algebra Lab	1			*		*
9	AME 5152	Applied Probability and Statistics Lab	1	*		*		*
10	BDE 5151	Algorithms and Data Structures for Big Data Lab	1			*		*
11	MPT 5100	Mini Project – I	4	*	*	*	*	*
12	PSD 5100	Professional Skill Development – I	1	*	*			
13	AME 5202	Deep Learning	3	*		*	*	
14	AME 5204	Applied Computer Vision	3			*	*	
15	AME 5205	Advanced Generative AI and Agentic Systems	3	*		*	*	
16	AME 5232	Reinforcement Learning	3	*			*	
	BDE 5001	Architecture of Big Data Systems	3	*			*	
	CCE 5001	DevOps for Cloud	3			*	*	*
	ESE 5234	Entrepreneurship	3			*	*	
17	AME 5241	Applied Mathematics for Machine Learning	3			*	*	
	AME 5243	Advanced Applications of Probability and Statistics	3	*		*		
	AME 5244	AI/ML on Cloud	3	*		*	*	
	ESE 5244	IT Project Management	3			*	*	
18	AME 5252	Deep Learning Lab	1	*				*
19	AME 5254	Applied Computer Vision Lab	1			*	*	*
20	AME 5255	Advanced Generative AI and Agentic Systems Lab	1			*	*	*

Sl. No.	Course Code	Course Name	Credits	PO1	PO2	PO3	PO4	PO5
21	AME 5282	Reinforcement Learning Lab	1				*	*
	BDE 5051	Architecture of Big Data Systems Lab	1			*	*	*
	CCE 5051	DevOps for Cloud Lab	1			*	*	*
	ESE 5284	Entrepreneurship Lab	1			*	*	
22	AME 5291	Applied Mathematics for Machine Learning Lab	1			*		*
	AME 5293	Advanced Applications of Probability and Statistics Lab	1			*		*
	AME 5294	AI/ML on Cloud Lab	1			*	*	*
	ESE 5294	IT Project Management Lab	1			*		*
23	MPT 5200	Mini Project – II	4	*	*	*	*	*
24	PSD 5200	Professional Skill Development – II	1			*	*	*
25	AME 6098	Project Work	30	*	*	*	*	*